Webpage Identity: College of Information Technology Education

Refer to: https://cite.nvsu.edu.ph/

Name: Career Path Recommendation System

Consent Form

By proceeding, I acknowledge that I am voluntarily providing personal and professional information for the purpose of assessing and predicting my potential career path. I understand that the results are for guidance purpose only and do not guarantee specific career outcomes.

Prediction Page (Students)

Course

- BSIT – Network Design Management
- BSIT – Web and Mobile Application Development
- BSCS – Game Development
- BSCS – Robotics
- BSIS – Business Analytics
- BSIS – Service Management

Academic Performance

GWA

- Outstanding (1.00 – 1.25)
- Very Satisfactory (1.26 – 2.00)
- Satisfactory (2.01 – 2.75)
- Fair (2.76 – 3.00)

Certifications

- Do you have any

Competencies (Based on CHED GTS)

- Technical Skills
  - Programming Skills
  - Data Analysis & Business Intelligence
  - Network Administration
  - Software Development
  - Technical Writing

- Cybersecurity
- Database Management
- Machine Learning & Artificial Intelligence

- Non-Technical Skills
  - Communication skills
  - Human Relations skills
  - Entrepreneurial skills
  - Information Technology skills
  - Problem-solving skills
  - Critical Thinking skills

RIASEC-based Interest Assessment

Use a 5-point Likert scale:

1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

REALISTIC (R) - Hands-on, technical, implementation-focused work

1. I prefer roles where I can directly configure, deploy, or maintain IT systems.
2. I enjoy working with actual hardware, networks, servers, or system environments.
3. I feel confident troubleshooting real-world technical problems independently.
4. I prefer operational or technical roles over purely theoretical or managerial ones.
5. I am comfortable working on-site or in environments that require hands-on system work.

Commonly aligned careers: Network Administrator, Systems Engineer, IT Support Specialist

INVESTIGATIVE (I) - Analytical, problem-solving, research-oriented

1. I enjoy solving complex programming, algorithmic, or system design problems.
2. I like analyzing system performance, security issues, or data patterns.
3. I am motivated to continually learn new technologies, languages, or frameworks.
4. I prefer roles that involve deep thinking, analysis, and logical reasoning.
5. I enjoy debugging, optimization, and finding root causes of system issues.

Commonly aligned careers: Software Developer, Data Analyst, Systems Architect

ARTISTIC (A) - Creative, innovative, Design-oriented

1. I enjoy designing user interfaces, user experiences, or interactive systems.
2. I like finding creative solutions to technical or user problems.
3. I enjoy projects that allow flexibility and innovation rather than rigid procedures.
4. I am interested in roles such as UI/UX design, game development, or multimedia systems.

5. I prefer work that combines creativity with technology.

Commonly aligned careers: UI/UX Designer, Front-End Developer, Game Developer

SOCIAL (S) - User-focused, collaborative, service-oriented

1. I enjoy working with users or clients to understand their needs.
2. I am comfortable explaining technical concepts to non-technical people.
3. I prefer team-based work rather than working alone all the time.
4. I feel satisfied when my work directly helps people or organizations.
5. I am interested in roles that involve training, coordination, or user support.

Commonly aligned careers: System Analyst, IT Trainer, IT Support, Business Analyst

ENTERPRISING (E) - Leadership, decision-making, business, and management orientation

1. I am interested in leading IT projects or teams.
2. I enjoy making decisions that impact organizational processes or systems.
3. I am confident in presenting ideas or solutions to stakeholders or management.
4. I am interested in the business side of IT solutions and digital transformation.
5. I see myself in roles involving management, consultancy, or entrepreneurship.

Commonly aligned careers: IT Project Manager, IT Consultant, Systems Manager

CONVENTIONAL (C) - Structured, detail-oriented, data and process-focused

1. I enjoy working with structured data, databases, or system documentation.
2. I prefer clear procedures, standards, and organized workflows.
3. I am comfortable managing records, reports, and system configurations.
4. I pay close attention to detail and accuracy in my work.
5. I prefer roles with defined responsibilities and systematic tasks.

Commonly aligned careers: Database Administrator, Systems Administrator, QA Analyst

Career Clarity

What are the 3 roles you are currently considering?

**Encoded:

First Option:_____

Second Option: ___

Third Option: ____

A. Career Fit Rating (Personality & Strengths)

Rate how well each role matches your personality and strengths

3 Roles

Scale:

1 – Very Poor Match

2 – Poor Match

3 – Fair

4 – Good Match

5 – Excellent Match

B. Growth & Learning Excitement Rating

How excited are you about learning and growing in each role?

- Show role choices given

Scale:

1 – Not Excited

2 – Slightly Excited

3 – Moderately Excited

4 – Very Excited

5 – Extremely Excited

C. Career Choice Motivation Scale.

Indicate the following how true each statement is for you.

1. I am genuinely interested in the type of work this career involves
2. This career is popular / high-paying / in demand
3. This career is expected by family or others

Scale:

1 – Strongly Disagree

2 – Disagree

3 – Neutral

4 – Agree

5 – Strongly Agree

D. Salary-Neutral Career Choice

1. If salary and job availability were equal, which career would you choose?
    - Choices based on Top 3 desired roles

E. RIASEC Alignment

Automatic Input based on RIASEC Assessment

Computation of Career Clarity

Career Clarity = A + B + C + D + E

Scoring Rubric

Component	Calculations	Max Score
Career Personality Fit	Average Score	5
Growth Excitement	Average Score	5
Genuine Interest	Average Score	5
Salary Neutral Consistency	If same to Top 1, 5, otherwise, 1.	5
RIASEC Alignment	If aligned sa result ng RIASEC sa desired choice = 5. Otherwise 1.	5

Maximum Score: 25

Total Score	Decision Confidence Level	Total Score
21–25	Very High Career Clarity	21–25
16–20	High Career Clarity	16–20
11–15	Moderate / Needs Guidance	11–15
6–10	Low Career Clarity	6–10
0–5	Highly Uncertain	0–5

Expected Output

- Top 3 Careers (Prediction)
- Top 3 Desired Roles
- Top 3 Interests
- Assessment based on Academic Performance, Interest, Competencies, and Career Clarity in alignment with their course.

For Administrator's Page

- Username & Password
- Train Model
- Accuracy
- Confusion Matrix